

NITIN VISHNOI

Senior ADAS & Embedded Software Engineer • 18+ Years • IEEE Senior Member (2025)
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PROFESSIONAL SUMMARY

Accomplished Senior Embedded Software Engineer with 18+ years delivering production-grade, safety-critical automotive software for General Motors, Stellantis, and Dura Automotive. Expert in AUTOSAR Adaptive/Classic BSW/ASW architecture, ISO 26262 functional safety (ASIL A–D), SOTIF (ISO 21448), and ASPICE Level 2+ process execution. Hands-on mastery of QNX Neutrino RTOS, Linux kernel BSP, NXP S32K/S32G SoCs, DDS middleware, CAN FD, Automotive Ethernet, TSN, and embedded cryptography (HSE/HSM, AES-CMAC, Secure Boot). Deep ADAS domain knowledge spanning sensor fusion (camera/radar/LiDAR/ultrasonic), L2+/L3 feature integration (ACC, LKA, AEB, FCW, APA, BSD, DMS), and Software-Defined Vehicle (SDV) zonal E/E architecture. IEEE Senior Member. Published SAE & ijeret author. DBA candidate with active research zonal SDV systems.

TECHNICAL SKILLS & CAPABILITIES

Languages	Embedded C (MISRA C:2012 compliant) • C++ (MISRA C++:2008, C++14/17) • Python 3 • Bash/Shell scripting • MATLAB • Assembly (ARM Cortex-R/M)
RTOS & OS	QNX Neutrino RTOS (IPC, Pulse/Signal, Resource Managers, Adaptive Partitioning) • Linux Kernel 5.x (BSP, Device Drivers, Yocto/Buildroot) • AUTOSAR OS (OSEK/OS-based) • FreeRTOS • Android AOSP BSP
AUTOSAR	AUTOSAR Adaptive (AP): ara::com, ara::diag, ara::exec, ara::idsm, Manifest-based deployment, SOME/IP, DDS/ROS2 • AUTOSAR Classic (CP): COM, PDUR, CANIF, DCM, DEM, FIM, ECUM, WDGm, NVM, MEMIF, FLASHIF, RTE, SWC design • Tools: Vector DaVinci Developer/Configurator Pro, EB Tresos Studio.
Safety & Security	ISO 26262 (ASIL A/B/C/D), FMEA, DFA, ASIL decomposition, Safety Case • SOTIF (ISO 21448) • ASPICE Level 2+ (SWE.1–SWE.6, MAN.3) • Secure Boot (HSM/HSE firmware, NXP S32K388 HSE) • AES-128/256, AES-CMAC, PKI, RSA, ECC, Key Wrap (RFC 3394) • TLS 1.3, SecOC, MAC verification
Protocols & Bus	CAN / CAN FD (ISO 11898) • Automotive Ethernet (100BASE-T1, 1000BASE-T1) • TSN (IEEE 802.1AS/Qbv/Qav) • SOME/IP • DDS (Data Distribution Service, RTI Connext / eProsima) • LIN 2.x • FlexRay • UDS (ISO 14229) / OBD-II • SPI / I2C / UART / SENT / PSi5 • DoIP (ISO 13400)
Hardware Platforms	NXP S32K3xx / S32G2/3 (Arm Cortex-M7/A55) • Renesas R-Car H3/M3 (Cortex-A57/A53) • Infineon AURIX TC3xx (TriCore) • NVIDIA Jetson Orin NX 16 GB • NVIDIA Jetson Nano Super • Raspberry Pi 4 / CM4 • Yahboom ROSMASTER R2 UGV • TI TDA4VM • STM32 (F4/H7)
ADAS / Perception	Sensor Fusion (Extended Kalman Filter, Unscented KF, Particle Filter) • Camera (ISP pipeline, V4L2, MIPI-CSi2, GStreamer) • Radar (object list / raw ADC, CAN/Ethernet interfaces) • Ultrasonic (PDC, APA parking grid) • ADAS features: ACC, LKA/LKAS, AEB, FCW, BSD, LCA, APA, DMS, TSR • OpenCV, TensorRT, ONNX Runtime, CUDA • ROS2 (Humble/Iron) — nodes, topics, services, actions, launch
Toolchain & DevOps	Vector CANoe / CANalyzer / CANdb++ / vFlash • Lauterbach TRACE32 / iSystem debugger • ETAS INCA / ISOLAR • dSPACE TargetLink / SCALEXIO (HIL) • LDRA / PC-lint / Polyspace (static analysis) • Git / Gerrit / GitHub Actions • Jenkins CI/CD • JIRA / Confluence / Polarion ALM • MATLAB Simulink / Embedded Coder • GCC / GHS Multi / Tasking / WindRiver Compiler
AI / ML	Vision Transformers (ViT, Swin-T) • CNN (ResNet, MobileNetV3, YOLO v8) • Reinforcement Learning (PPO, DQN — zonal workload scheduling) • RAG (Retrieval-Augmented Generation for ADAS decision fusion) • TensorFlow / PyTorch / TensorRT • ONNX export & quantization (INT8/FP16) • Edge inference optimization (Jetson Orin NX, Jetson Nano Super)
Standards & Process	ISO 26262 • ISO 21448 (SOTIF) • ASPICE (Automotive SPICE) • ISO/SAE 21434 (Cybersecurity) • UN ECE WP.29 R155/R156 • AUTOSAR (Classic 4.x / Adaptive 22-11) • MISRA C:2012 / MISRA C++:2008 • SAE J3016 (Autonomy Levels) • Agile/SCRUM • V-Model SDLC

PROFESSIONAL EXPERIENCE

Senior Embedded Software Engineer

General Motors (GM) • Milford, MI, USA

May 2023 – Present

- SDV Platform Architecture (Zonal E/E + SOA):
 - Working on Next-Gen ECU SDV- software development
 - Architect and develop next-generation ECU software for software-defined vehicles with a focus on real-time performance, security, and scalability.
 - Lead middleware and low-level software stack implementation across multiple automotive ECU platforms.
 - Design and integrate AUTOSAR-based frameworks enabling reliable vehicle control, diagnostics, and deterministic real-time processing.
 - Implement automotive cybersecurity solutions, including secure boot, cryptographic services, and key provisioning.
 - Optimize QNX/Linux BSPs for multi-core automotive ECUs, improving system stability and performance.
 - Designed zonal ECU software topology replacing domain controllers — implemented SOA inter-zone communication via SOME/IP and AUTOSAR Adaptive aram::com, reducing harness weight by ~15% (program estimate).
- Embedded Cybersecurity (HSE / HSM Firmware):
 - Implemented MacGen/MacVerify LLSIs on NXP S32K388 HSE using pre-allocated job descriptor pools, per-job state machines, lock-free MU channel bitmap allocation, and one-pass AES-CMAC (<4 KB data path).
 - Engineered AES-256 key export (plaintext + RFC 3394 AES KeyWrap); designed async interrupt-driven multi-channel HSE queue with full test suite achieving <2 μ s MAC latency overhead.

ADAS Platform Technical Lead

KPIT Technologies — Client: Stellantis (FCA) • Auburn Hills, MI, USA

Feb 2021 – May 2023

- ADAS Feature Integration (L2+ / L3):
 - Led SW integration of ACC (host-vehicle velocity/gap control), LKA (lateral deviation with lane departure warning, LCA (radar rear-quadrant coverage) on AUTOSAR Classic ECUs.
 - Defined sensor abstraction interfaces for Continental ARS4 radar (CAN FD object list), Mobileye EyeQ5 camera CSI2 feed, and Bosch ultrasonic ring — achieving <5 ms sensor-to-controller latency budget compliance.
 - Led ADAS platform architecture and integration for Level 2+ and Level 3 features, including ACC, LKA, APA, and Automated Parking.
 - Designed and deployed AUTOSAR middleware solutions enabling real-time sensor fusion and perception stacks.
 - Ensured ISO 26262 functional safety and ASPICE Level 2+ compliance through close collaboration with systems and validation teams.
 - Developed and optimized middleware supporting radar, ultrasonic, and camera sensors, improving perception reliability and vehicle safety.
 - Acted as the primary technical interface between OEM stakeholders, system architects, and development teams.

ADAS Software Engineer → Manager

Dura Automotive Systems • Auburn Hills, MI, USA

Oct 2018 – Jan 2021

- Custom ADAS Middleware Stack (QNX / MISRA C++):
 - Built ground-up ADAS middleware on QNX Neutrino — message-passing IPC channels, resource managers for sensor I/O, priority-ceiling mutexes for shared object lists; worst-case response time <10 ms on safety path.
 - Delivered SW modules for LKAS (polynomial lane model, PID lateral correction), AEB (TTC-based braking arbiter), FCW (kinematic TTI threshold), DMS (face-landmark drowsiness classifier).
- DDS Middleware & Sensor Fusion Pipeline:
 - Designed DDS-governed multi-agent communication (eProsima Fast-DDS) across ADAS domain controllers — 18 sequence diagrams covering 17 ADAS functions; enforced DEADLINE QoS policies (ACC: 20 ms, AEB: 10 ms, DMS: 50 ms).
 - Implemented Extended Kalman Filter (EKF) fusion module merging radar and camera tracks into a unified object list; validated against CARLA simulation data with <0.3 m positional RMSE.
- Multi-Sensor Fusion Architecture:
 - Integrated 6x Continental radar (CAN FD ARS4 object list), 12x ultrasonic (LIN 2.2 Valeo), 5x cameras (MIPI-CSI2 + V4L2, OpenCV pre-processing); fused tracks via IMM-EKF algorithm.
 - Designed sensor arbitration logic handling sensor degradation modes (missing, frozen, out-of-range) with ASIL-B safe-state fallback — improving overall perception availability by 28%.
- Team Leadership & Process:
 - Led and mentored 10+ engineers; introduced MISRA C++ coding standard, peer-review gates, and Polyspace static analysis — driving a 40% reduction in escape defects over 12 months.
 - Established Agile sprint cadence (2-week sprints, JIRA Kanban board) aligned with ASPICE SWE work products, improving delivery predictability to 92% sprint-commitment rate.

Technical Specialist → Technical Manager

Capgemini — Client: Denso USA • Southfield, MI, USA

Apr 2014 – Oct 2018

- QNX Cluster & HUD Systems (Production — Honda / GM vehicles):

- Led full-lifecycle development of QNX Neutrino-based instrument cluster and HUD — OpenGL ES 2.0 rendering pipeline, MOST-150 and CAN gateway driver, ISO 26262 ASIL-B display path.
- Managed 18-engineer team across SW development, system integration, validation, and OEM PPAP release; delivered 3 major programs on schedule with zero ITAR escapes.
- Established RTOS performance benchmarking (IRQ latency profiling, context-switch histograms using Lauterbach TRACE32) — practices subsequently adopted as BU standard.

Technical Lead — Embedded R&D

HCL Technologies • USA

Nov 2010 – Apr 2014

- Developed Android BSPs (AOSP 4.x) for ARM Cortex-A9 tablets and smart appliances; brought up SPI, I2C, and USB OTG drivers for custom sensor peripherals.
- Designed embedded firmware (C, ARM assembly) for a production microwave system integrating 50+ automated recipes via NFC tag provisioning and I2C keypad scanning.

Software Engineer → Senior Software Engineer

SPA Computers Ltd • Bangalore, India

Dec 2007 – Nov 2010

- Developed automotive display firmware (AUTOSAR Classic CAN/LIN BSW) and GPS/GNSS tracking firmware with camera integration; optimized ISR routines achieving 40% MCU efficiency gain.
- Designed and validated CAN, UART, SPI, and I2C drivers compliant with AUTOSAR MCAL layer specifications; improved bus throughput by 25% through DMA-assisted transfer configuration.

RESEARCH PUBLICATIONS & INTELLECTUAL CONTRIBUTIONS

- **Vishnoi, N.** (2026). A FIVE-ERA TAXONOMY AND BENCHMARK FRAMEWORK FOR LANE DETECTION: FROM CLASSICAL HEURISTICS TO VISION FOUNDATION MODELS IN AUTONOMOUS DRIVING [Journal-article]. International Journal of Artificial Intelligence and Applications (IJAIA), 17(2), 95–98. <https://aircconline.com/ijaia/V17N2/17226ijaia06.pdf>
- **Vishnoi, N.,** Indrakanti, R. K. K., & P, V. K. V. S. R. (2026). AI-Driven ADAS in Software-Defined Vehicles: architectures, safety assurance, and lifecycle challenges. ijeret.org. <https://ijeret.org/index.php/ijeret/article/view/449>
- Indrakanti, R. K. K., **Vishnoi, N.,** & Kamadi, V. (2026). A System-Level calibration framework for embedded vision: integration of sensor, firmware, and software enhancements. SAE Technical Papers on CD-ROM/SAE Technical Paper Series, 1. <https://doi.org/10.4271/2026-01-0013>
- P, V. K. V. S. R., Indrakanti, R. K. K., **Vishnoi, N.,** & Chitta, V. (2026). Leveraging trusted platform modules for hardware rooted security and robust device encryption. ijeret.org. <https://ijeret.org/index.php/ijeret/article/view/450>

EDUCATION

Doctor of Business Administration (DBA) — Autonomous Driving & Intelligent Systems

ESGCI Paris, France | Expected to complete by 2027

Dissertation: Unified Zonal Architecture Framework (UZAF) | RQs: Zonal Topology Design, AI Compute Placement, SW Platform Selection (Adaptive AUTOSAR vs QNX vs Linux)

Master of Science(2 year) — Machine Learning & Artificial Intelligence

Liverpool John Moores University | 2023

Executive PGP(1 Year) — Artificial Intelligence & Machine Learning | IIIT Bangalore | Dec 2022

Diploma(1 Year) — Embedded System Design (VLSI, MCU, RTOS) | CDAC Hyderabad | Dec 2007

Bachelor of Technology(B.Tech-4 year degree) — Electronics & Communication Engineering | MIT Moradabad | Dec 2005

CERTIFICATIONS & RECOGNITION

- IEEE Senior Member — Elevated 2025 • SAE International Member
- U.S. EB-2 NIW Permanent Residency Approved — Nationally Significant Contributions in AI, Embedded Systems & Automotive Software
- AUTOSAR in Practice — Vector North America (Classic & Adaptive)
- Functional Safety (ISO 26262) — Automotive Software Safety Engineering
- Automotive Cybersecurity — Secure Boot, HSM/HSE & Cryptographic Services
- Self-Driving Car Engineer Nanodegree — Udacity (Sensor Fusion, Localization, Path Planning, Control)
- Flying Car & Autonomous Flight Engineer — Udacity (Motion Planning, Estimation, Controls)
- Software Product Management Specialization — University of Alberta